Point Calibration Does Not Control Type I Error Across the Composite Null in a Composed Adaptive Bayesian Design

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Abstract

Efficacy thresholds for Bayesian monitoring can be calibrated at a single assumed control rate. We examined whether such point calibration controls Type I error (T1E) across the composite null in a two-arm Phase II design with historical control borrowing, Bayesian monitoring, sample size re-estimation (SSR), and response-adaptive randomization (RAR). Each component passed its numerical benchmark check, but none of those checks evaluated T1E away from the calibration anchor.

The full-design threshold was calibrated to $\alpha = 0.025$ at $p_0 = 0.25$ with no time trend. We then held the threshold fixed and varied only the common baseline rate over $p = 0.20$–0.45, with no trend, so that every evaluated point is a constant-rate null and every rejection is a Type I error. SSR used the blinded pooled response rate only as a nuisance-parameter estimate while holding the planned treatment effect fixed.

At the calibration anchor the design performed as intended: estimated Type I error 0.0236 (MCSE 0.0021) against a nominal 0.025. Changing only the baseline rate raised it to 0.0810 at $p = 0.30$, 0.1854 at $p = 0.36$ and 0.2786 at $p = 0.45$, roughly eleven times nominal, with the curve still climbing at the grid boundary. No calendar trend and no misspecified analysis are required to produce this failure. A separate trend-active grid, in which both arms drift together under a time-constant analysis, is reported as a stress analysis; its rejections are not Type I errors and reach 0.2874. At the same fixed threshold, disabling SSR and RAR did not restore Type I error control on the constant-rate grid either; that ablation reached 0.2020 at $p = 0.45$. The fixed-threshold difference between the full design and the ablation reversed sign across the range. The SSR × RAR interaction was estimated twice; the inverse-variance pooled estimate is -0.0077 (95% simulation interval $[-0.0157, +0.0002]$), which does not support an interaction claim in either direction.

The grid demonstrates a control failure but does not locate the supremum. Operating characteristics should therefore be evaluated across prespecified null-generating conditions, not only at the calibration anchor.

1. Introduction

Computational tools are used to specify priors, simulate operating characteristics and prepare regulatory submissions. In the workflow examined here, the checks applied to such a tool are component-level and numerical: each adaptive mechanism is compared against a reference implementation in isolation. This paper asks what those checks do and do not establish about the error rate of the assembled design. We make no claim about how widespread that workflow is; the question is what follows from it where it is used.

The FDA's January 2026 draft guidance on the use of Bayesian methodology in clinical trials [U.S. Food and Drug Administration, 2026] is directly relevant. The guidance recommends that sponsors document operating characteristics under realistic simulation scenarios, pre-specify how prior-data conflict will be handled, and evaluate whether Bayesian adaptive designs meet frequentist operating characteristic targets. (It is a draft guidance containing nonbinding recommendations, and we describe it as such throughout.) These recommendations bear on a specific practice, and one this paper examines directly: an efficacy threshold calibrated at a single assumed control rate, with that single point then treated as evidence of error control.

This paper makes two claims, at two levels. The primary claim is about **robustness**: an efficacy threshold calibrated at one assumed control rate does not deliver Type I error control across the composite null. The secondary claim is about **composition**: at the frozen full-design threshold, the combined SSR/RAR contrast changes sign across the evaluated null rates, but the factorial interaction estimates are too imprecise to support a stable interaction claim in either direction.

Both claims are demonstrated by simulation over a grid of common baseline rates. We are explicit about what such a grid can and cannot establish: it exhibits points at which control fails, but it does not locate or estimate the supremum of the Type I error over the null.

We present Zetyra as the reproducible evaluation infrastructure for this work. The platform is evidence and infrastructure for the argument; it is not the argument itself.

2. Background

2.1 What Component-Level Checks Establish

"Validation" of clinical trial design software is not a term with a single agreed meaning. Where it amounts to documenting that design outputs were computed with a named package, with cross-checks against published values, it does not establish maximum numerical deviation across the range of design parameters, does not guarantee reproducibility under stochastic methods, and does not account for the behaviour of multiple adaptive components operating jointly. The FDA January 2026 draft guidance raises the standard explicitly: operating characteristics should be evaluated by comprehensive simulation over plausible assumptions—the guidance discusses prior-data conflict explicitly and calls for scenarios spanning plausible values of design inputs, including the background rate of the endpoint—and documented reproducibly.

The simulations and component benchmarks are implemented in Zetyra. The versioned source, package information and seeds that produced every reported number are released with the paper (see Acknowledgements); the public validation repository contains the broader software checks.

2.2 Composed Adaptive Designs and Marginal Calibration

Definition. A *composed adaptive design* is a design in which two or more adaptive mechanisms read the same interim data before the final analysis. In the design studied here, Bayesian monitoring, the SSR trigger and the RAR allocation rule all compute from the accumulated responses at interim looks.

We do not formalise this as a set-theoretic condition on information sets. Writing $\mathcal{F}_i \cap \mathcal{F}_j \neq \emptyset$ would not distinguish anything: for sigma-fields the intersection always contains $\emptyset$ and Ω, so the condition holds trivially for any pair. Nor is the converse available— informational independence does not by itself guarantee that error control survives adaptation.

The relevant observation is weaker and purely negative. Let $\phi(X)$ denote the final rejection rule

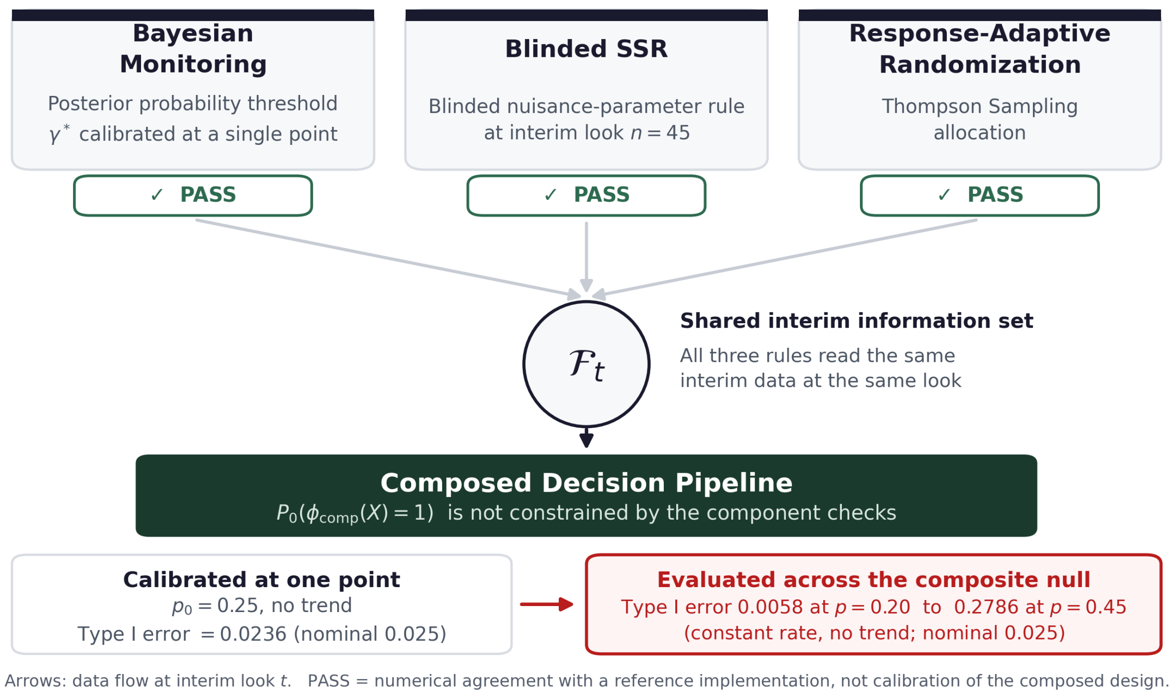

Figure 1: Composed adaptive design: Bayesian monitoring, blinded nuisance-parameter SSR and RAR share interim data before the final analysis. Component checks and point calibration at the anchor do not bound Type I error elsewhere in the composite null.

(Bayesian monitoring), and let $A_1(X_t)$, $A_2(X_t)$ denote the adaptation rules—SSR and RAR—which alter sample size or allocation based on interim data X_t but do not themselves reject H_0. Each is validated independently (the blinded nuisance-parameter SSR satisfies its pooled-input, algebra and bound checks; RAR tracks $P(\theta_e > \theta_c)$ within tolerance). The composed decision rule is $\phi_{\text{comp}}(X) = \phi\big(X;\, A_1(X_t),\, A_2(X_t)\big)$: the final rejection indicator evaluated on data whose sample size and allocation were shaped by the adaptation rules. The standard approach calibrates that composed rule at a single point, choosing γ^* so that $P_{p_0,\,\delta=0}\big(\phi_{\text{comp}}(X) = 1\big) \approx \alpha$ at one anchor rate with no trend. That establishes the error rate at that point and nowhere else: it places no bound on $P_{p,\,\delta}\big(\phi_{\text{comp}}(X) = 1\big)$ for other (p, δ) in the null. The component benchmark checks add no such bound either, since none of them evaluates an error rate.

Calibrating each component under its own marginal model, and the composed rule at a single null point, places no constraint on the error rate of the composed rule elsewhere in the null. This is a statement about what those checks fail to establish, not a condition under which inflation must occur. Whether it occurs, in which direction and by how much are empirical questions, which is why the remainder of the paper is a simulation study. Figure 1 illustrates the structure.

3. Simulation Study

3.1 Design Specification

We simulate a two-arm Phase II oncology design with MAP borrowing on the control arm [Weber et al., 2021], interim looks at $n = 30, 45, 60$ and a final at $n \geq 90$ (post-SSR). RBesT fits the historical cohorts ($n = 50, 40, 60$; 38/150 responses) as a robustified mixture with components $0.5887\,\mathrm{Beta}(18.403, 53.273)$, $0.2113\,\mathrm{Beta}(1.956, 4.703)$ and $0.2000\,\mathrm{Beta}(1, 1)$. At 50% enrollment, Rule B uses only the pooled response count to estimate the overall event-rate nuisance parameter, while the planned treatment effect $\Delta = 0.45 - 0.25 = 0.20$ remains fixed. Under a prespecified working 1:1 allocation, it preserves the fixed-sample Wald power implied by the original $N = 90$ design (0.5296), permits increases only, and applies the protocol cap $N = 150$. The working rates are $\hat{p}_c = \mathrm{clamp}(\hat{p}_{\mathrm{pool}} - \Delta/2)$ and $\hat{p}_e = \hat{p}_c + \Delta$; the recalculated per-arm size is

$$n_{\mathrm{arm}} = \frac{(z_{1-\alpha} + z_{\mathrm{power}})^2[\hat{p}_c(1 - \hat{p}_c) + \hat{p}_e(1 - \hat{p}_e)]}{\Delta^2},$$

rounded up and doubled. This pooled-data construction follows the blinded binary internal-pilot principle of Friede and Kieser [2004]. It is a conventional-style working nuisance rule, not an exact power calculation for the Thompson-sampling allocation or Bayesian final decision. At the planned $n = 45$ look, its complete count mapping spans only $N = 90$–100; the exact expected N is 90.44 at $p = 0.25$, 91.57 at $p = 0.30$, and 94.12 at $p = 0.36$ under the active trend. Thompson Sampling RAR is benchmarked against analytical $P(\theta_e > \theta_c)$.

The threshold is calibrated at the design anchor $p_0 = 0.25$, distinct from the dominant-component mean 0.2568, robustified-mixture mean 0.3132 and historical pooled rate 0.2533. We therefore describe $p > 0.25$ only as departure from the calibration anchor, not from the prior mean.

Under H_0, both arms share baseline rate p and drift together, so the reported rejection probabilities are frequentist Type I error, not posterior predictive error. We evaluate $p = 0.27$ (mild) and $p = 0.30$ (strong) with linear trend $\delta = 0.05$ over the planned enrollment horizon. These are sensitivity scenarios, not an asserted empirical range; Tang et al. [2010] and Viele et al. [2014] provide clinical and historical-borrowing context.

All estimates carry binomial MCSE $\sqrt{\hat{p}(1 - \hat{p})/N_{\mathrm{sim}}}$; cell-specific MCSEs and 95% simulation intervals are reported here or in the simulation supplement.

Repeated nominal conditions use independent seeds and simulation budgets; their largest difference is about 2.0 combined standard errors. We report each in its originating experiment rather than pooling it.

Note on calibration: The efficacy threshold is obtained by grid-free binary search, giving $\gamma^* = 0.96566$ with in-sample T1E = 0.0248 (MCSE 0.0011, N = 20,000) at the design anchor $p_0 = 0.25$ with no trend. On an independent fresh-seed sample (N = 5,000), T1E was 0.0224 (MCSE 0.0021; 95% simulation interval $[0.0183, 0.0265]$), compatible with the nominal target. Binary search still targets an empirical sample rather than the population error rate, so neither estimate is exact calibration.

3.2 Component-Level Benchmark Checks

Each adaptive component is individually benchmarked. MAP prior: maximum deviation from RBesT mixture = 1.78×10^{-15} (floating-point precision). Rule B SSR: the calculation reproduces $N = 90$ at the exact planning nuisance rate, uses only the pooled interim count, is increase-only, returns even sample sizes, and stays within the prespecified bounds; its full $n = 45$ input support was exhaustively checked. The pooled nuisance construction is compared with the framework of Friede and Kieser [2004], but the embedded working Wald rule is not claimed to reproduce the operating characteristics of their final test. RAR one-step allocation: maximum deviation from analytical $P(\theta_e > \theta_c) = 0.0036$ (Monte Carlo error at n = 100,000). All three components pass their benchmark checks; none of those checks evaluates Type I error away from the anchor.

3.3 Type I Error Across the Composite Null

We evaluate a one-dimensional slice of the composite null, fixing $\delta = 0.05$ and varying the common baseline rate p. Both arms share the same response process, so H_0 holds throughout. No row is the calibration scenario: even the $p = 0.25$ row has an active trend, whereas γ^* was tuned at $\delta = 0$.

3.4 The Constant-Rate Composite Null

We first hold $\delta = 0$, so the response rate is constant over calendar time and identical in both arms at every point of the grid. Every point is then a genuine null for the analysis actually performed, and every rejection is a Type I error in the conventional sense. This isolates the robustness question from any question about model misspecification.

p	SSR + RAR disabled	All mechanisms	MCSE (full)	Fixed-γ^* difference
0.20	0.0298	0.0058	0.0011	−0.0240
0.22	0.0356	0.0090	0.0013	−0.0266
0.25	0.0534	0.0236	0.0021	−0.0298
0.27	0.0780	0.0380	0.0027	−0.0400
0.30	0.0936	0.0810	0.0039	−0.0126
0.33	0.1330	0.1314	0.0048	−0.0016
0.36	0.1562	0.1854	0.0055	+0.0292
0.40	0.1778	0.2358	0.0060	+0.0580
0.45	0.2020	0.2786	0.0063	+0.0766

At the calibration anchor $p = 0.25$ the full design attains 0.0236 (MCSE 0.0021), consistent with the nominal 0.025 and confirming that the calibration itself is sound. Moving the common baseline rate alone raises Type I error monotonically: 0.0380 at $p = 0.27$, 0.0810 at $p = 0.30$, 0.1854 at $p = 0.36$ and 0.2786 at $p = 0.45$. **Calibration at a point does not deliver control over the composite null, and no calendar trend or misspecified analysis is needed to show it.**

The mixture prior has mean 0.3132, so the two largest rates characterise the shape of the curve rather than representing plausible control rates for this indication. We note also that $p = 0.45$ is a *null* scenario in which both arms sit at 0.45, so there is no treatment effect; the value is numerically equal to the experimental-arm rate assumed under the planning alternative (against a 0.25 control), but the coincidence is arithmetic only and that row is not an alternative-hypothesis evaluation. Over the clinically motivated portion of the grid—departures of two to eleven percentage points from the anchor—Type I error runs from 0.0380 to 0.1854, up to 7.4 times nominal.

3.5 Trend-Active Stress Analysis

We next activate a shared calendar trend in both arms while retaining the time-constant analysis. Contemporaneously there is still no treatment effect, so this remains a no-effect scenario, but the analysis is now misspecified with respect to the data-generating process. Rejections here are therefore *not* Type I errors in the conventional sense, and we report them as rejection probabilities throughout.

p	SSR + RAR disabled	All mechanisms	Fixed-γ^* difference
0.20	0.0342	0.0092	−0.0250
0.22	0.0474	0.0208	−0.0266
0.25	0.0742	0.0484	−0.0258
0.27	0.0900	0.0754	−0.0146
0.30	0.1180	0.1206	+0.0026
0.33	0.1398	0.1670	+0.0272
0.36	0.1644	0.2286	+0.0642
0.40	0.1894	0.2798	+0.0904
0.45	0.1970	0.2874	+0.0904

Two readings follow. First, **at the full-design threshold, disabling SSR and RAR does not return the rejection probability to the nominal level**: with both switched off, it still rises from 0.0342 at $p = 0.20$ to 0.1970 at $p = 0.45$. That column is the same fixed-threshold ablation used throughout (Section 3.6)—it carries γ^*, which was calibrated with SSR and RAR active. At $p = 0.25$ on this trend-active slice it is 0.0742; neither that row nor the full-design row is the no-trend calibration scenario. Read it as "the design with the adaptive components switched off, threshold unchanged", not as a calibrated monitoring-only design. Second, **at fixed γ^*, enabling SSR and RAR lowers the rejection probability at low p and raises it at high p**: the difference runs from −0.0250 at $p = 0.20$ to +0.0904 at $p = 0.45$, reversing sign across the range.

The grid establishes a direction change, not its crossing point: the contrast at $p = 0.30$ is +0.0026 against a Monte Carlo standard error of 0.0065, and the neighbouring rates are 0.27 and 0.33. It also shows elevated rejection probabilities across the grid but does not locate their supremum. The largest *estimated* rejection probability on this grid is 0.2874; that is a Monte Carlo estimate carrying an MCSE of 0.0064, not a deterministic bound, and a one-sided 95% simulation limit places that rate's rejection probability at or above 0.2769.

Comparing this grid with the constant-rate grid of Section 3.4 at matched baseline rates, the trend's contribution is *not* monotone in the departure. It rises across most of the range—+0.0034 at $p = 0.20$, +0.0248 at $p = 0.25$, +0.0396 at $p = 0.30$, +0.0440 at $p = 0.40$—and falls to +0.0088 only at $p = 0.45$, where it is indistinguishable from zero (1.0 SE). The fall from $p = 0.40$ to $p = 0.45$ is −0.0352, about 2.8 standard errors, and rests on one point at the grid boundary. We report it as an observation about that point and offer no mechanism. Over the range evaluated here the trend-active scenario is the more severe one everywhere except at the largest rate, where the two are indistinguishable; it should not be described as uniformly worse, nor as converging. The more aggressive Rule A stress test is summarized in Appendix A.

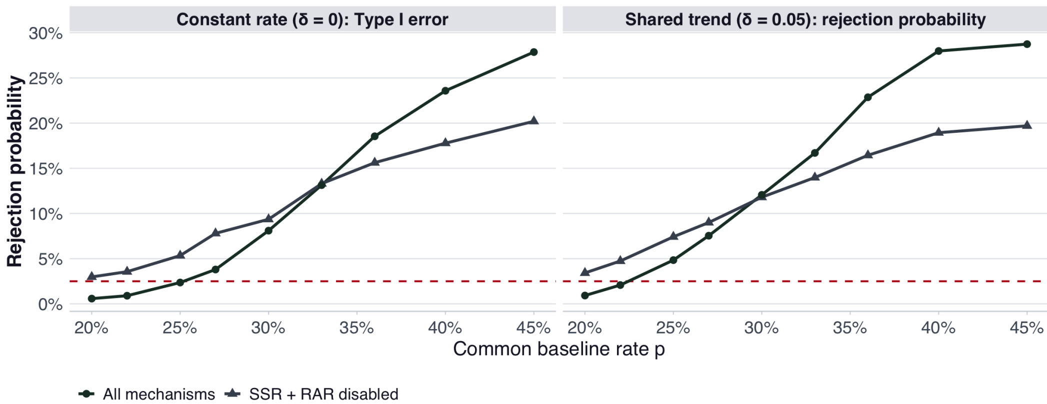

Figure 2: Rejection probability across the common-rate grid. **Left panel:** the constant-rate composite null ($\delta = 0$), where the analysis is correctly specified and every rejection is a Type I error. **Right panel:** the trend-active stress scenario ($\delta = 0.05$), where both arms drift together but the analysis remains time-constant; those rejections are *not* Type I errors and are reported as rejection probabilities. In both panels, grey holds the full-design threshold with SSR and RAR disabled and dark green shows all mechanisms; both are fixed-threshold comparisons, not separately calibrated designs. Dashed red line: nominal $\alpha = 0.025$.

3.6 Fixed-Threshold Factorial Ablation: SSR by RAR

Four designs (SSR $\times$ RAR), all at $p = 0.30$ with the trend active:

Condition	Rejection probability	MCSE
SSR + RAR disabled	0.1120	0.0045
+ SSR only	0.1248	0.0047
+ RAR only	0.1236	0.0047
+ SSR and RAR	0.1124	0.0045

These are ablations at a fixed threshold, not a decomposition. γ^* was calibrated with SSR and RAR active; disabling components while retaining it does not yield separately calibrated designs or attributable shares. The baseline cell is therefore not "the excess caused by monitoring", although it does show that disabling both components leaves the rejection probability far above the nominal level. These cells are trend active, so the quantity elevated here is a rejection probability, not a Type I error.

The SSR-enabled cells also accrue an additional interim analysis. The monitoring schedule is $\{30, 45, 60, 90\}$ together with the realised final sample size, so a trial that SSR extends beyond $N = 90$ is analysed five times while a fixed-N trial is analysed four. Because the test statistic is a maximum over looks, the SSR-enabled cells receive one further opportunity to cross the threshold in the $\Pr(N > 90)$ fraction of replicates, which is 0.3157 at $p = 0.30$. This is correct trial conduct – an extended trial does analyse at its realised final N – but it means every full-design-versus-ablation contrast reported here combines the adaptive mechanisms with a difference in multiplicity, and no contrast should be read as isolating the mechanisms alone.

The simple contrasts are $\Delta_{\text{SSR}|\text{RAR}=0} = +0.0128$ and $\Delta_{\text{RAR}|\text{SSR}=0} = +0.0116$: each mechanism alone raises the rejection probability by roughly one percentage point, while enabling both raises it by +0.0004.

The interaction was estimated twice, independently. The first run ($N = 5{,}000$ per cell) gave -0.0240 (propagated MCSE 0.0091, 95% SI $[-0.0419, -0.0061]$), which excludes zero. A higher-precision replicate on a disjoint seed block ($N = 20{,}000$ per cell) gave -0.0037 (MCSE 0.0046; 95% SI $[-0.0126, +0.0052]$), which does not. The two are 2.0 standard errors apart (heterogeneity $Q = 3.96$ on 1 df, $p = 0.047$) – near the edge of what independent sampling produces unaided, and not on its own evidence of a defect. Their inverse-variance pooled estimate is $\Delta_{\text{interaction}} = -0.0077$ (SE 0.0041; 95% SI $[-0.0157, +0.0002]$).

We report the pooled value and draw no directional conclusion. The lower-precision run's interval excluding zero is not reproduced by the better-powered replicate, and we do not quote it on its own. Joint evaluation remains necessary because an interaction is possible, but its sign and magnitude should not be inferred from these runs.

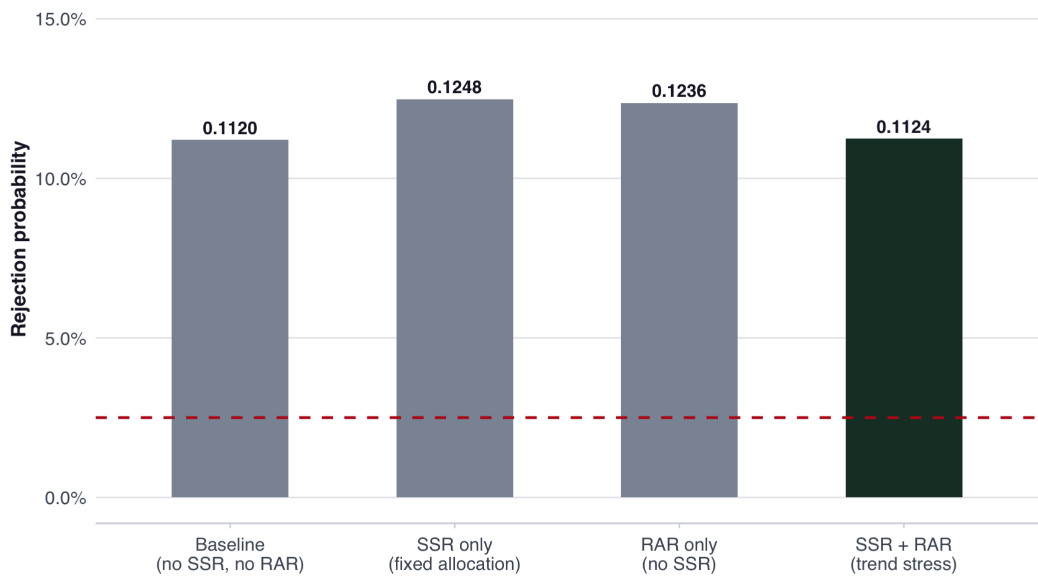

Figure 3: Fixed-threshold Rule B factorial ablation of SSR by RAR at $p = 0.30$ with the shared trend active. Simple contrasts $\Delta_{\text{SSR}} = +0.0128$ and $\Delta_{\text{RAR}} = +0.0116$. The interaction is estimated twice; the inverse-variance pooled estimate is -0.0077 (95% SI $[-0.0157, +0.0002]$), which includes zero. Bar labels are cell estimates; the interaction is not annotated in the panel because it is a four-cell contrast rather than a property of any one bar. Dashed red line = nominal $\alpha = 0.025$.

3.7 Baseline-Rate Departure by Time Trend

A 2×2 factor crossing evaluates departure and trend under the primary design; both factors feed monitoring, SSR and RAR, so it does not isolate those channels.

Condition	Rejection probability	vs. calibration
Neither (calibration)	0.0232	—
Departure only	0.0904	+0.0672
Trend only	0.0474	+0.0242
Both	0.1140	+0.0908
Additive prediction	0.1146	
Interaction term	−0.0006	

The departure-only simple contrast is +0.0672, and the trend-only simple contrast is +0.0242. Because both factors feed monitoring, SSR and RAR, neither contrast is attributable to a single channel. Unlike the fixed-threshold ablations, all four cells share the full design and a common calibrated baseline. The interaction is −0.0006 (MCSE 0.0071; 95% SI $[-0.0145, +0.0133]$) and is indistinguishable from zero: the joint effect of a baseline departure and a calendar trend is, at these factor levels, what adding the two separate effects would predict. It is a risk-difference interaction at these factor levels and includes the additional trend exposure from enrollment beyond $N = 90$; Rule B limits that extension to 100 at the planned SSR look.

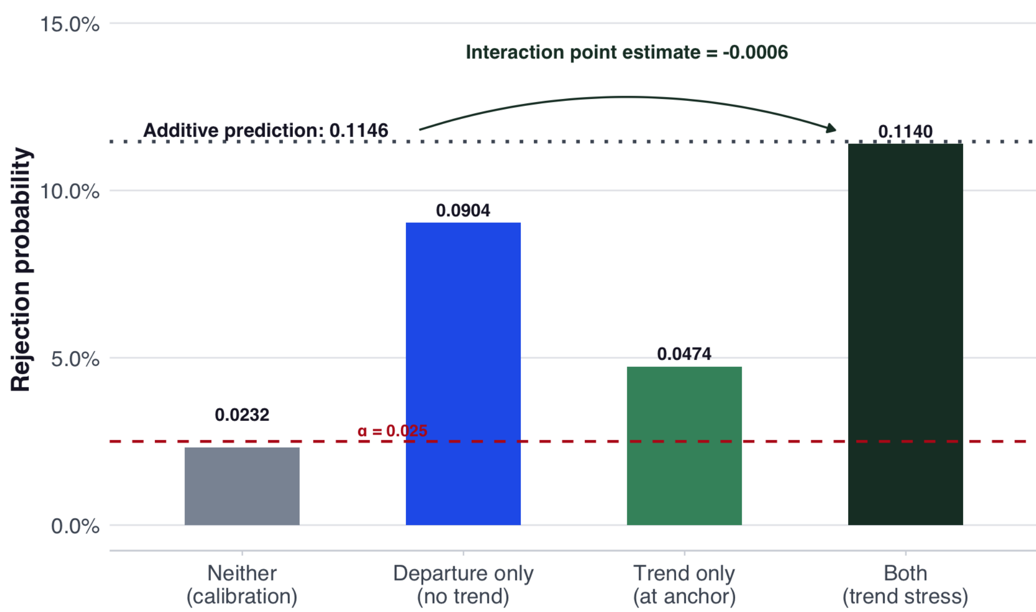

Figure 4: Departure-by-trend crossing under the full design. The interaction is −0.0006 (95% SI $[-0.0145, +0.0133]$); dashed red line denotes nominal $\alpha = 0.025$.

3.8 Sensitivity Analyses

The following sweeps use the strong departure parameterisation ($p = 0.30$) to characterize mechanism sensitivity across a wider range.

- **Departure severity**, at constant trend ($p \in \{0.25, 0.27, 0.30\}$, all with $\delta = 0.05$): the rejection probability increases monotonically, $0.0484 \to 0.0754 \to 0.1206$, with no direction reversal.
- **Vague-component weight** (0–90%): larger weight on the flat robustify component means less borrowing; the rejection probability declines from 0.1488 to 0.0624.

- **Time trend** δ (0–15%): the rejection probability increases from 0.0812 at $\delta = 0$—which is a Type I error, the trend being absent—to 0.2220 at $\delta = 0.15$.
- **SSR trigger timing**: moving the primary look from $n = 45$ to $n = 30$ changed the rejection probability from 0.1161 to 0.1244; a separate sweep over $n_{\mathrm{SSR}} \in \{20, 30, 40, 45, 55, 60\}$ ranged from 0.1126 to 0.1234. Timing also changes the realised final-N distribution, so these are not isolated timing effects.

4. Discussion

4.1 Regulatory Implications and Recommendations

An efficacy threshold calibrated at one assumed control rate need not control Type I error elsewhere in the composite null. Consistent with the FDA January 2026 draft guidance, operating-characteristic evaluation should therefore span realistic null-generating conditions and pre-specify how prior–data conflict will be handled.

For this design class, sponsors should (1) evaluate a grid of plausible control rates rather than one anchor; (2) label whether component ablations are separately calibrated, since a fixed-threshold ablation tests restoration of control rather than attribution; and (3) cross departure and time trend rather than rely only on separate one-factor analyses.

4.2 Mitigation Strategies

Potential safeguards include more robust or dynamic borrowing, pre-specified prior–data disagreement diagnostics, pipeline-level recalibration of γ^* over the null grid, and an explicit maximum acceptable T1E over a stated robustness range. Each changes the composed pipeline and must itself be evaluated at pipeline level; no component-level argument guarantees restored control.

4.3 Limitations and Future Work

This paper demonstrates a control failure by simulation in a specific design class; several limitations bound what it establishes.

The supremum is not estimated. The grid exhibits points at which control fails. The largest *estimated* Type I error on the constant-rate grid is 0.2786, which is a Monte Carlo estimate with an MCSE of 0.0063 and may lie either side of the true value at that rate; a one-sided 95% simulation limit places it at or above 0.2682. Neither figure locates the worst case, and the curve has not flattened at the grid boundary—the final three increments are +0.0540, +0.0504 and +0.0428—so the supremum over the composite null is plausibly well above any value evaluated here. A probabilistic bound on the supremum itself would require a search over the null, not a finite grid.

Conflict is not quantified. We vary the calibration-anchor departure, not a prior-predictive conflict measure. Extending prior–data diagnostics such as Matthews and Qian [2026], Qian and Matthews [2026] from Normal-mean settings to this binary MAP-mixture design would require a new statistic and calibration.

Rule B is a working nuisance calculation, not exact design power. Its 1:1 Wald model differs from the RAR allocation and Bayesian final decision; the 0.5296 reference power and 90–100 mapping follow from the original $N = 90$ design. Results are conditional on this prespecified rule. Appendix A summarizes the more aggressive Rule A stress test.

No sufficient condition is proved. Shared information neither guarantees nor precludes error control; a conditional-error or design-specific argument would be required.

Calibration is Monte Carlo. Binary search targets the empirical calibration sample; a fresh-seed validation was compatible with nominal, but neither finite simulation establishes the population error rate exactly.

Component ablations are not separately calibrated. They retain the full-design γ^* and answer whether disabling a component restores control, not how much excess it caused. Attribution would require calibrating each design separately.

5. Conclusion

Point calibration did not provide uniform Type I error control across the composite null. At the full-design threshold, disabling SSR and RAR did not restore it on the constant-rate grid. The fixed-γ^* contrast reversed sign across the evaluated null rates, but the Rule B factorial and departure-by-trend interaction intervals both included zero; the simulations do not support a stable interaction claim in either direction.

The practical consequence is a reporting standard rather than a new method. Operating characteristics should be evaluated over a prespecified grid of null-generating conditions. Fixed-threshold ablations can show whether disabling a component restores control, but attribution requires separately calibrating each design. The simulation code is publicly available for independent verification and extension; the Zetyra platform is a commercial SaaS product.

Calibrate at a point. Validate over the null.

Acknowledgements

Simulation source, manuscript source, figures and reproducibility instructions: github.com/evidenceinthewild/Composed-Adaptive-Trial-Validation. Rule A is summarized in Appendix A as a method-development stress test, not as the primary analysis.

A. Original Rule A Pooled Conditional-Power Stress Test

The original Rule A treated the pooled interim rate as a one-sample signal against $p_0 = 0.25$ and increased total sample size to target 80% conditional power, capped at $N = 150$. A common upward nuisance shift under the two-arm null could therefore drive expansion, so it was replaced before publication by Rule B. With $\gamma^* = 0.96596$, Rule A reached a rejection probability of 0.2618 at $p = 0.36$ under the trend-active null grid, compared with 0.2286 under Rule B at the same rate. This more aggressive result is retained only as a stress-test summary; full outputs and source are not part of the primary Rule B release.

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